3.3 Leverage-augmented VAR (LVAR)

This extension asks whether commodity variance histories add forecast information after accounting for negative equity returns. We compare a plain VAR with a *leverage-augmented VAR* (LVAR). The relevant benchmark for commodity additions is an equity-only model with the same variance lags and downside controls. This separates the contribution of equity-return information from that of additional markets.

Let z_t contain ES log RV5 and the included commodity log variances. Our LVAR is the linear VAR with predetermined explanatory variables (VAR-X)

$$z_{t+1} = c + \sum_{j=1}^{p} A_j z_{t+1-j} + \Gamma d_t + u_{t+1}, \qquad p = 5. \tag{1}$$

Each equation is estimated by OLS. The plain VAR sets $\Gamma = 0$; the equity-only versions are the corresponding AR and AR-X benchmarks. Neither Treasury proxies nor calendar indicators enter this extension, distinguishing it from the seasonally controlled VAR in Section 3.2. The term "leverage" refers here to the predictive association of negative equity returns with variance; it does not identify a particular economic mechanism or a regime-switching process.

Motivated by persistent leverage effects (Corsi and Renò, 2012), the three controls are defined for decimal ES close-to-close log returns r_t by

$$d_t = (d_t^{(1)}, d_t^{(5)}, d_t^{(22)})', \qquad d_t^{(h)} = \left| \min\left(\frac{1}{h} \sum_{i=0}^{h-1} r_{t-i}, 0 \right) \right|. \tag{2}$$

The stored percentage returns are divided by 100. Importantly, the negative part is taken *after averaging signed returns*; the weekly and monthly terms are not averages of daily negative parts. These controls enter every system equation and are available at the forecast origin. The specification is our application of a multihorizon downside-control idea, rather than a replication of the full LHAR-CJ model in Corsi and Renò (2012).

The plain VAR follows the standard reduced-form framework (Lütkepohl, 2005, Chapters 2–4). For every included market, both VAR and LVAR retain all five variance lags. The plain VAR in this subsection is the matched benchmark for isolating the contribution of the downside controls. It is distinct from the specification in Section 3.2, which includes calendar controls and selects four lags in a system containing Treasury variation. A direct comparison with that model would therefore change several features at once and would not isolate the leverage contribution.

The final panel contains 3,910 observations from 4 January 2011 to 31 August 2026. Shared HAR/LHAR warm-ups leave 3,867 regression targets, including 3,084 initial training targets through 12 July 2023. BIC on the initial five-market *plain* VAR selects five lags from candidates 1–20, using identical training outcomes for every candidate; AIC selects ten. The five-lag order is held fixed across the main plain and leverage comparisons. The 1/5/22 return windows are fixed features, not separately selected VAR orders. Fits expand chronologically to produce 783 forecasts for the next retained observation, from 13 July 2023 to 31 August 2026. No target-date commodity values or returns enter the forecast.

Table 1: Jake's matched forecast comparison on the final dataset

Model	ES coefficients	MSE	MAE
Equity-only AR(5)	6	0.368531	0.464175
Equity-only AR(5) + downside controls	9	0.334215	0.442837
LVAR: oil and gold	19	0.336611	0.443919
LVAR: natural gas	14	0.332670	0.442343
LVAR: oil, natural gas and gold	24	0.334549	0.443412
LVAR: all four commodities	29	0.334687	0.443561

Notes: identical 783 ES log-RV5 outcomes and forecast origins. Lower loss is better. Coefficients include the intercept and refer to the ES equation, not the whole system. "All four" means oil, corn, gold and natural gas. Source: own calculations in `VAR_QTFE.ipynb`, final-data run of 13 October 2026.

Adding downside controls to the equity-only AR(5) reduces forecast MSE by 9.31%. Adding oil and gold to the resulting equity-only LVAR instead increases MSE by 0.72%; adding all four commodities increases it by 0.14%. Natural gas alone lowers MSE by 0.46%. Descriptive paired circular-block intervals for these commodity comparisons include zero, whereas the downside-control improvement remains positive across the examined block lengths. The largest clearly observed gain in this extension therefore comes from equity-return information rather than a broad commodity history.

In-sample joint HAC tests still find conditional commodity associations, illustrating why historical fit and incremental forecast performance must be distinguished. The results support modelling equity-return asymmetry within this comparison, while providing little evidence of a useful incremental commodity forecast gain. The small commodity differences do not establish statistically significant forecast superiority or the absence of economic links between markets.

ADF rejects a unit root for all five training log-RV5 series, while KPSS rejects level stationarity. We retain log-variance levels but treat this disagreement as an unresolved specification limitation. Conditional companion eigenvalues are below one in the reported training fits, yet ES residual dependence remains. OLS inference uses HAC covariance with bandwidth 22 (Newey and West, 1987); this does not resolve nonstationarity or omitted dynamics. Ten-lag sensitivity lowers equity-only LVAR MSE to 0.328510, with oil and gold still failing to improve its matched benchmark. Forecast rankings and uncertainty intervals remain exploratory because the evaluation period has already been inspected. We make no claim of structural causality or tested gains in portfolio utility or Value-at-Risk.

Appendix J: figures for the final-data results

This appendix is optional supporting material. For a short main report, keep Table 1 and use Figure 1 to show incremental information and uncertainty; move the other charts to the empirical appendix. Figure 2 is an alternative to a detailed accuracy table. All figures below are generated from verified final-data outputs.

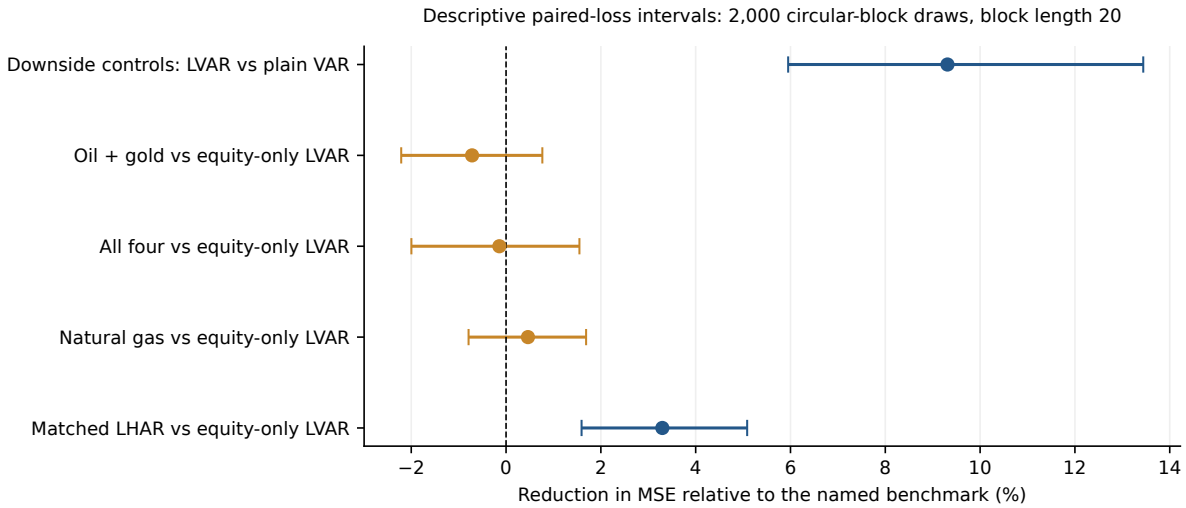


Figure 1: Incremental information in forecasts of ES log RV5. Points show percentage reductions in MSE relative to each named benchmark; values above zero favour the candidate. Bars are descriptive 95% paired circular-block intervals based on 2,000 draws and block length 20, using the same 783 forecast targets. They are not formal nested-model or selection-adjusted significance tests. Commodity intervals span zero; the downside-control gain is much larger. The LHAR row is a separate common-date benchmark calculation, not the original HAR notebook's default-split score.

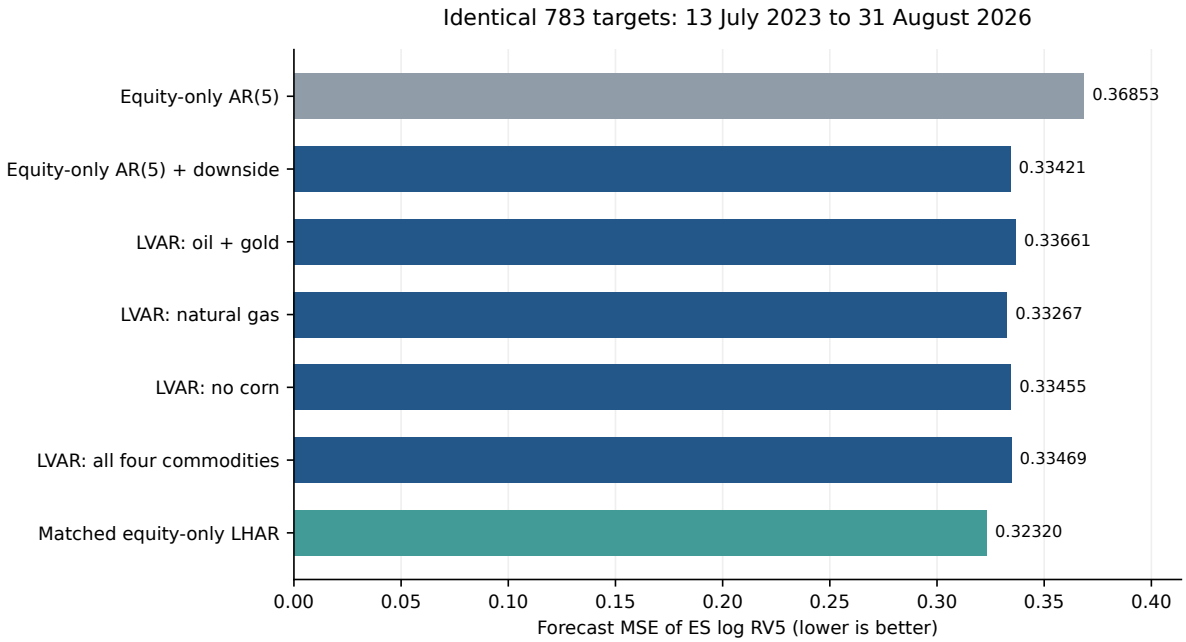


Figure 2: Forecast accuracy with matched equity-only benchmarks. Bars report MSE of ES log RV5 over the same 783 targets; the axis begins at zero. Every VAR/LVAR retains all five lags of its included markets; LVAR additionally includes the three downside controls. The LHAR reference is recomputed on these common dates. Differences from Tim's VAR or the original HAR table also reflect specification and estimation-sample choices, so they cannot be attributed to leverage alone.

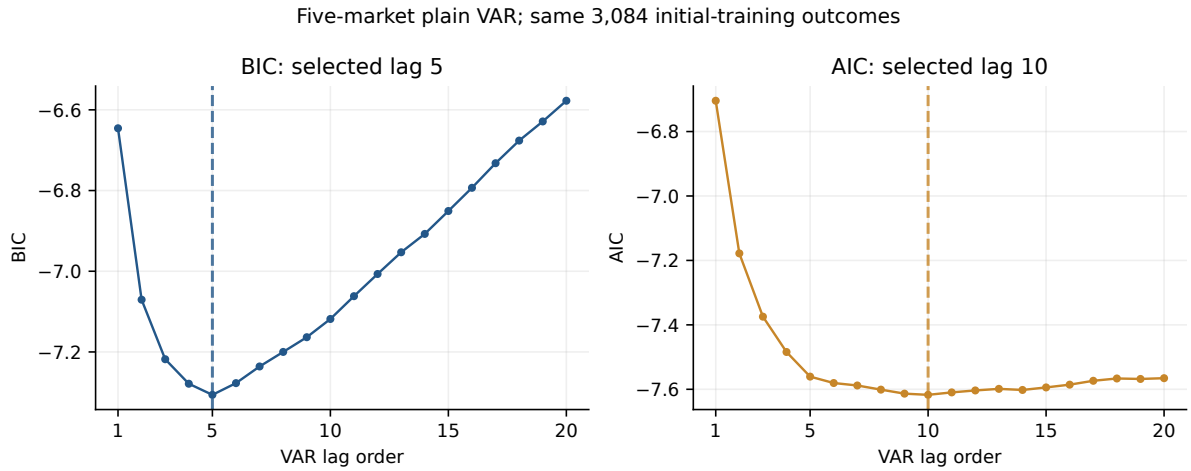


Figure 3: Initial-training VAR lag selection. The plain five-market system has an intercept and log-RV5 lags, with no downside, Treasury or seasonal controls. Each candidate uses the same 3,084 training outcomes. BIC selects five lags and AIC ten; these are separate selection criteria, not stationarity tests. These curves belong to Jake's specification and should not replace Tim's six-variable seasonally controlled selection figure.

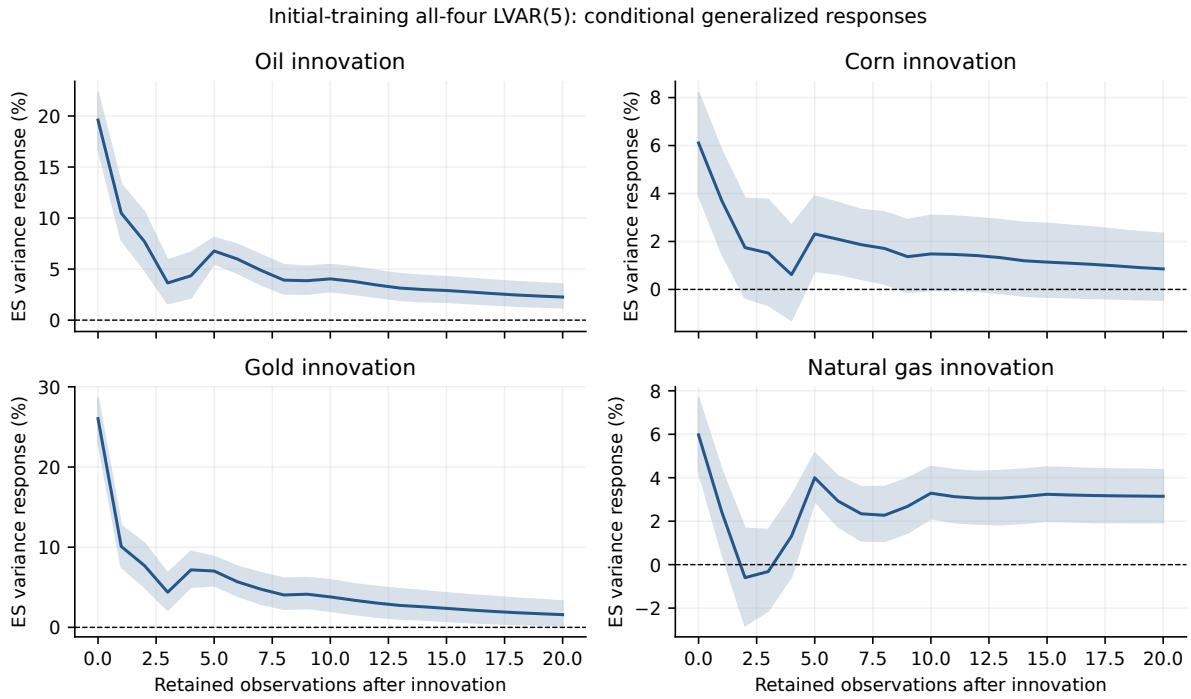


Figure 4: Conditional generalized commodity-to-equity responses in the initial-training all-four LVAR(5), using the generalized-response framework of Pesaran and Shin (1998). Each innovation is one residual standard deviation of commodity log RV5, including contemporaneously correlated innovations. The same predetermined downside-control path is used in the shocked and baseline systems. Responses are displayed as $100[\exp(g_h) - 1]$ percent changes in variance; $h = 0$ is contemporaneous. Shading is an exploratory 95% pointwise joint-HAC parameter simulation band, conditional on admissible draws. These are conditional model responses, not identified causal shocks or forecasts of future commodity prices.

Integration notes: align the group report before submission

These notes are for the authors and should be removed from the final report body. The complete review is in `reports/final_dataset/GROUP_REPORT_REVIEW.md`. Teammates' code and the supplied group PDF have not been changed.

Stationarity and model checks. ADF/KPSS appropriately test the individual data series before estimation. They involve their own auxiliary calculations; they are not tests of a fitted VAR. Fitted companion eigenvalues, residual autocorrelation and ARCH checks are complementary. No fitted model can be used to certify away conflicting series-level stationarity evidence.

Lag conventions. The implementation choices are:

Model or check	Current rule
Jake VAR/LVAR	Training-only five-market BIC: 5 lags; AIC: 10. Fixed across main comparisons.
Tim VAR	Training-only six-variable system plus month/weekday controls: BIC 4, AIC 9.
HAR/LHAR	Fixed means over 1, 5 and 22 observations, each entering as one regressor.
ADF/KPSS	Jake: AIC autolag/automatic bandwidth; Tim: fixed 20/28. These are test settings.
HAC(22)	Covariance bandwidth, not a VAR order.

Dates and data. The final CSV starts on 4 January 2011, not 5 January. It has complete corn data; corn enters the updated Tim and Jake VAR extensions. ES close returns are used in LHAR and LVAR. The supplied yield proxies describe changes between retained yield observations, rather than necessarily consecutive native Treasury dates. The baseline models do not all impose an extra Treasury availability delay; Tim reports that as a sensitivity. Avoid a blanket delayed-yield claim in the shared data section.

Comparable losses. Tim and Jake score the same 783 outcomes, but Tim's initial training history has 3,127 raw rows, while Jake's common warm-ups leave 3,084 training targets. Original HAR uses a separate 80/20 split with 778 test targets beginning 20 July 2023; LHAR has 774 beginning 26 July. Their printed scores must not be presented as the same-date comparison. Current original baseline MSEs are 0.3639 and 0.3254, respectively; the older group PDF shows 0.3592 and 0.3213 and needs its HAR table refreshed.

Interpretation and structure. Replace the report's mean-of-daily-negative return formula with Equation (2). Distinguish the downside gain from incremental commodity value and from comparisons across differently controlled models. A percent change in log-variance MSE does not imply the same percent change in VaR. Repair the contents bookmarks, remove duplicated economic interpretation and comments, complete the empty results/conclusion sections, and attach the referenced appendices. Tim's current BIC results round to AR MSE 0.356690 and an all-commodity MSE increase of 0.64%.

Report naming. Keep Section 3.2 as "Vector autoregression (VAR)" for Tim's calendar/Treasury specification and Section 3.3 as "Leverage-augmented VAR (LVAR)" for Jake's return-control extension. Give the basic VAR equation once, then state each extension's controls, lag order and training warm-up. Table labels such as "AR(4), calendar controls," "AR(5)," and "AR(5) + downside controls" identify the matched benchmarks. Interpret the 9.31% gain against Jake's own plain benchmark, rather than Tim's differently controlled model.

Figure priorities. Use the incremental-gain chart in the main text if the accuracy table is retained. The accuracy chart can replace the large table. Lag selection and generalized IRFs belong in the appendix; IRFs are optional if space is tight. The notebook's residual grid and stationarity table can support the limitations discussion, rather than being presented as proof of a fully adequate model. The 5/10 sensitivity chart is useful alongside the lag table.

References

Corsi, F., and Renò, R. (2012). Discrete-time volatility forecasting with persistent leverage effect and the link with continuous-time volatility modeling. *Journal of Business & Economic Statistics*, 30(3), 368–380. doi:10.1080/07350015.2012.663261.

Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer. Chapters 2–4 on stable VAR processes, estimation, order selection and model adequacy. doi:10.1007/978-3-540-27752-1.

Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610.

Pesaran, H. H., and Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1), 17–29. doi:10.1016/S0165-1765(97)00214-0.